

Carpenter High Permeability "49"

Type Analysis:

Carbon.....0.05%
Manganese.....0.05%
Silicon.....0.35%
Nickel.....48.0%
Iron.....Balance

DESCRIPTION:

Carpenter High Permeability "49" is 48% nickel-iron alloy, which has a saturation flux density of approximately 15,000 gauss after hydrogen annealing. Hysteresis loss is very low in either DC or AC circuits in which the frequency is less than 400 cycles per second. It is used in laminated cores for instrument transformers, magnetic shields and cores for certain electronic and communication devices in which extremely high permeability at low magnetizing forces greatly increases the efficiency and effectiveness of the equipment. Because of its high permeability, it is used in solenoid cores in light, sensitive relays that must operate and respond to weak currents that induce low magnetizing forces.

After it is given a low temperature hydrogen anneal, 1300/1400°F (704/706°C), High Permeability "49" exhibits constant permeability characteristics. Its initial permeability is approximately 1000 and it will remain close to this value at flux densities up to approximately 500 gauss.

Carpenter High Permeability "49" is available in three grades. Two of these - High Permeability "49" and High Permeability "49" Rotor Grade ♦ show minimum directional magnetic properties.

High Permeability "49" is available as bars, wire, rod and strip in thicknesses of 0.020" (0.05 cm) and over. Its major uses include magnetic shields and solid cores for solenoids and other control devices.

High Permeability "49" Rotor Grade is supplied as un-oriented strip for laminations 0.004/0.020"(0.010/0.051 cm) thick. It is usually used in rotating components such as resolvers and servo-synchros whose magnetic properties must be nondirectional.

Carpenter High Permeability "49" ♦ Transformer Grade is specially processed to produce higher permeabilities at low magnetizing forces in the directions parallel to, and perpendicular to the rolling direction. It is available in strip 0.004/0.020" (0.010/0.051 cm) thick. Its primary application is for transformer laminations.

PHYSICAL CONSTANTS:

Specific gravity.....8.250
Density ♦ lb/in♦.....0.298

Mean coefficient of thermal expansion

in/in/♦F x 10--6 (77/400♦F).....5.20
cm/cm/♦C x 10--6 (25/204♦C)..... 9.40

Electrical resistivity

ohm-cir mil/ft 290
microhm-cm 48

Temperature coefficient of electrical resistivity

per ♦F (0/930♦F) 0.002
per ♦C (-17.8/499♦C) 0.0036

Specific heat

BTU/lb/♦F 0.12
kcal/kg/♦C 0.12

Thermal conductivity

BTU/in/ft♦/hr/♦F 90.0
kcal/m/hr/♦C 11.2

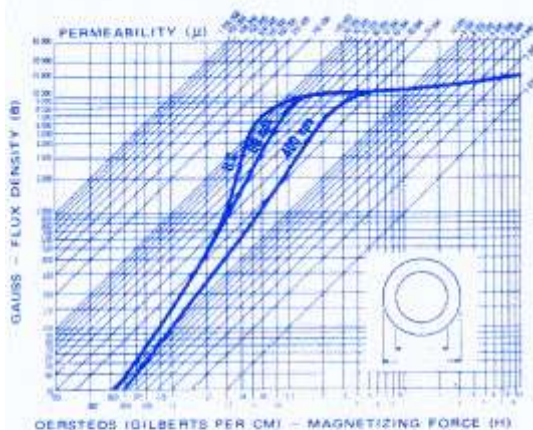
Curie temperature.....840/930♦F (449/499♦C)

Melting point.....2600♦F (1427♦C)

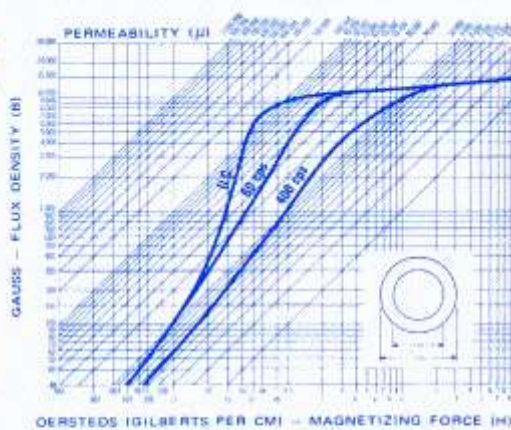
MAGNETIC PROPERTIES:

Grade	Thickness			Minimum 60 Cycle AC Permeability			
	inches	cm	B40	B200	B2000	B4000	B8000
Rotor	0.014	0.036	6,000	10,000	26,000	37,000	41,500
Transformer	0.014	0.036	10,000	14,500	32,000	40,000	45,000
Rotor	0.006	0.015	6,500	11,000	32,000	45,000	60,000
Transformer	0.006	0.015	11,000	17,000	46,000	56,000	62,000

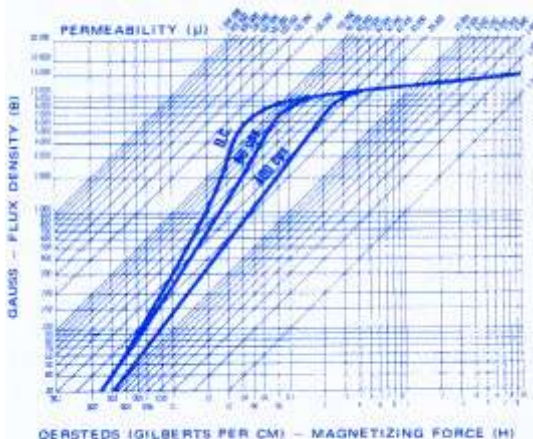
These minimum permeability requirements are based on evaluating the properties via a ring specimen 1♦" (3.81 cm) OD x 1" (2.54 cm) ID hydrogen annealed at 2150♦F (1177♦C) four hours and furnace cooled with an approximate cooling rate of 150/220♦F (83/122 C♦) per hour through the Curie Point.



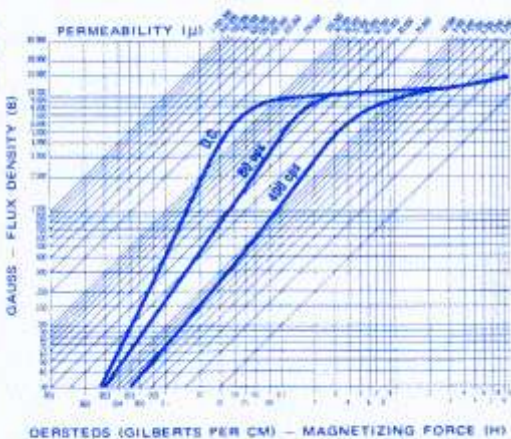
Carpenter High Permeability "49" – Rotor Grade – typical magnetization curves at DC, 60 cps and 400 cps from stamped ring specimen 0.006" (0.015 cm) thick dry hydrogen annealed at 2150° F (1177° C) for four hours.



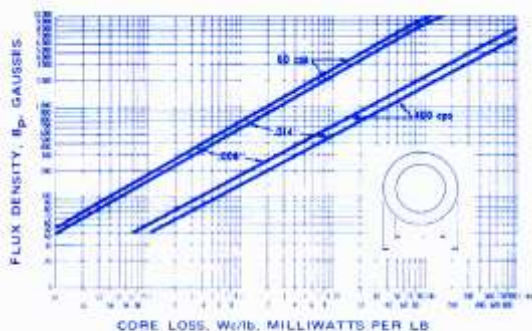
Carpenter High Permeability "49" – Rotor Grade – typical magnetization curves at DC, 60 cps and 400 cps from stamped ring specimen 0.014" (0.036 cm) thick dry hydrogen annealed at 2150° F (1177° C) for four hours.



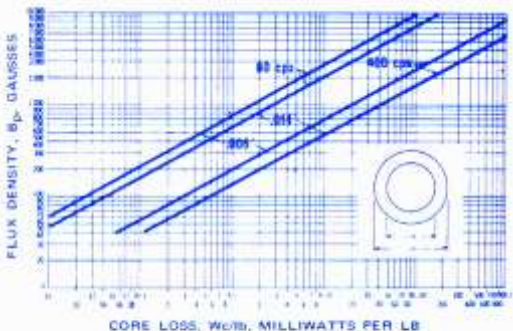
Carpenter High Permeability "49" – Transformer Grade – normal magnetization curves at DC, 60 cps and 400 cps sine flux excitation 1-DU laminations, 0.006" (0.015 cm) thick. Hydrogen annealed, 2150° F (1177° C), 4 hours.



Carpenter High Permeability "49" – Transformer Grade – normal magnetization curves at DC, 60 cps and 400 cps sine flux excitation 1-DU laminations, 0.014" (0.036 cm) thick. Hydrogen annealed, 2150° F (1177° C), 4 hours.



Carpenter High Permeability "49" – Rotor Grade – specific core loss curves at 60 cps and 400 cps from stamped ring specimen 1" (2.54 cm) ID x 1½" (3.81 cm) OD dry hydrogen annealed at 2150° F (1177° C) for four hours.



Carpenter High Permeability "49" – Transformer Grade – specific core loss curves at 60 cps and 400 cps from stamped ring specimen 1" (2.54 cm) ID x 1½" (3.81 cm) OD dry hydrogen annealed at 2150° F (1177° C) for four hours.

DC Magnetic Properties determined according to ASTM A-341.

	Bars	Strip 0.025" (0.064 cm) To 0.125" (0.318 cm)
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Standard Hydrogen Anneal: For maximum softness and optimum magnetic and electrical properties, High Permeability "49" should be annealed in an oxygen-free, dry hydrogen atmosphere with a dew point below -40°F (-40°C) for two to four hours at 2150°F (1177°C), followed with a furnace cool of 100/200°F (38/93°C) per hour down to 800°F (427°C) and any rate thereafter. Oil, grease, lacquer and all other contaminants must be removed before annealing. The

Initial Permeability B100	5,000	6,500
Maximum Permeability	50,000	60,000
Residual Magnetism, Gauss	6,000	9,000
Coercive Force, Oersteds from 10,000 Gauss	0.05/0.10	0.05/0.08
from 5,000 Gauss	0.02/0.03	0.02/0.03
Saturation Inductance from H-100 Oersteds, Gauss	15,000	15,000

Temperature Variation of Permeability
(Expressed in % of Value at 77°F (25°C)).

Temperature		Permeability at 1000 Gauss	Maximum Permeability
°F	°C		
-94	-70	75/90	90/95
212	100	120/130	100/105

AC and DC Magnetic Properties of lamination strip 0.006" (0.015 cm) and .014" (0.036 cm) thick.

HEAT TREATMENT:

Strain Relieving Anneal: To relieve all strains and restore the alloy to a soft condition suitable for drawing, spinning, forming, bending and similar operations, anneal for not more than one hour at 1450/1600°F (788/871°C). Since high nickel, high permeability alloys readily absorb carbon, sulfur, oxygen and other contaminants from combustion furnace gases, it is recommended that in-process annealing be done in a hydrogen or inert gas atmosphere.

individual parts should be separated by an inert insulating powder such as magnesium or aluminum oxide during hydrogen annealing or surface insulation media.

WORKABILITY:

Machinability: If components are to be machined in volume from bar stock, High Permeability "49-FM", a free machining grade, is recommended.

The regular grade, High Permeability "49", machines somewhat like austenitic stainless alloys. It develops gummy chips but does not work harden as rapidly as the stainless alloys. Sulfur-bearing cutting compounds are highly detrimental to the final magnetic properties. Animal lard oil is recommended for drilling and machining operations which must be done at slow speeds. Work hardened bars (Rockwell B 85/100 minimum) offer best machining characteristics. Parts should be degreased and cleaned as soon as possible.

Cold Forming: For best blanking characteristics, strip should be purchased in the cold rolled condition, Rockwell B 90 minimum. For best forming characteristics, strip should be purchased as cold rolled annealed for forming. Best drawing characteristics are obtained when strip is purchased as annealed, Deep Drawing Quality.

Forging: The recommended forging temperature for High Permeability "49" is 2150°F (1177°C).

Welding: It is readily brazed, welded, soft and hard soldered by following the usual practices for ferrous alloys.

CORROSION RESISTANCE:

High Permeability "49" resists weather and moisture corrosion to a moderate extent.

FORMS AVAILABLE:

Strip, cold rolled or annealed
Wire, cold drawn
Bars, hot rolled, cold drawn and centerless ground
Billets for forging

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